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An active learning variable-fidelity metamodelling approach based on ensemble of metamodels and objective-oriented sequential sampling

Qi Zhou, Xinyu Shao, Ping Jiang, Zhongmei Gao, Hui Zhou and Leshi Shu

The State Key Laboratory of Digital Manufacturing Equipment and Technology, School of Mechanical Science and Engineering, Huazhong University of Science & Technology, Wuhan, People' Republic of China

ABSTRACT

Computational simulation models with different fidelity have been widely used in complex systems design. However, running the high-fidelity (HF) simulation models tends to be very time-consuming, while incorporating low-fidelity (LF), inexpensive models into the design process may result in inaccurate design alternatives. To make a trade-off between high accuracy and low expense, an active learning variable-fidelity (VF) metamodelling approach aiming to integrate information from both LF and HF models is proposed. In the proposed VF metamodelling approach, a model fusion technology based on ensemble of metamodels is employed to map the difference between the HF and LF models. Furthermore, an active learning strategy based on a generalised objective-oriented sequential sampling strategy is introduced to make full use of the already-acquired information of difference characteristics between the HF and LF models. Several numerical and engineering cases verify the applicability of the proposed VF metamodelling approach. Different types of test cases, sample sizes, and metamodel performance evaluation measures including accuracy and robustness are considered.

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Variable fidelity; model fusion; metamodel; ensemble; active learning strategy

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1. Introduction

For complex system design, deterministic computer simulations are often used to replace the computation-intensive and controlled real-life experiments. However, running simulations for a complex system with multiple inputs and outputs can become computationally prohibitive as is the case of computational fluid dynamics (CFD) and finite element analysis (FEA). For example, it is reported that it takes Ford Motor Company about 36–160 hours to run one crash simulation for a full passenger car (Crombecq et al. 2011; Crombecq, Laermans, and Dhaene 2011). Indeed, it is still impractical to directly use this simulation with an optimiser to evaluate a lot of design alternatives when exploring the design space for an optimum (Aute et al. 2013; Zheng et al. 2014). This limitation can be addressed by adopting metamodel (or surrogate), which can mimic the original system at a considerably reduced computational cost (Gao et al. 2012). There are a lot of commonly used metamodels, such as Polynomial Response Surface models

CONTACT Ping Jiang jiangping@mail.hust.edu.cn
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(Li et al. 2008), Kriging models (Xiao et al. 2012), Artificial Neural Networks models (Schmidhuber 2015), Radial Basis Function (RBF) models (Fang and Horstemeyer 2006; Zhou et al. 2015a, 2015b), and Support Vector Regression (SVR) models (Clarke et al. 2005). A good review on various metamodelling techniques can be found in Wang and Shan (2007) and Forrester and Keane (2009). It is important to point out that the quality of the metamodels has a profound impact on the computational cost and convergence characteristics of the metamodel-based design optimisation. Generally, sample points from high-fidelity (HF) simulations and models offer more information of the system, however, at a higher cost (Shan and Wang 2010). Sample points from low-fidelity (LF) models require lower expense, while lead to inaccurate metamodels even distorted metamodels. Hence, conflict between high accuracy and low expense seems to be inevitable in building metamodels.

To ease this problem, variable-fidelity (VF) metamodelling approaches based on the interaction of HF and LF models have been widely concerned (Viana et al. 2014). Commonly used VF metamodelling approaches are scaling methods, which tune the LF model according to the response values of the HF model. These scaling methods can be divided into two distinct types: local VF metamodelling approaches and global VF metamodelling approaches. In local VF metamodelling approaches, the scaling function is approximated using local metamodels, for example, linear regression (Chang et al. 1993), first/second Taylor series (Alexandrov and Lewis 2001; Alexandrov et al. 2001; Eldred and Dunlavy 2006). The local VF metamodelling approaches are easy to be implemented and can achieve a relatively high accuracy within an appropriate trust region size, while they are only suitable for local optimisation problems (Gano, Renaud, and Sanders 2005, 2006; Zheng et al. 2015). In global VF metamodelling approaches, the scaling function is approximated using global metamodels, namely, Kriging scaling methods (Gano et al. 2006; Han, Görtz, and Zimmermann 2013; Zheng et al. 2013), RBF scaling methods (Van Nguyen, Tyan, and Lee 2014; Tyan, Nguyen, and Lee 2014), SVR scaling methods (Sun, Li, and Li 2012; Zheng et al. 2014), and Bayesian-Gaussian scaling methods (Forrester and Keane 2007; Xiong, Chen, and Tsui 2008). Compared with the local VF metamodelling approach, the most obvious advantage of these global VF metamodelling approaches is that they are able to cope with multiple optimum situations sophisticatedly on the entire domain (Zheng et al. 2013; Van Nguyen, Tyan, and Lee 2014). However, these practices may increase the risk of adopting an inappropriate metamodel that is used in lieu of the scaling function for VF metamodelling when considering the following two realities: (a) the nature of the scaling function is not a priori and the resource to obtain information describing the relationship between input variables and the response values of scaling function is limited; and (b) the accuracies of constructed metamodel for scaling function depend on the current training data, and a different metamodel type may become more accurate than the selected one with a new available data set.

Moreover, how to make full use of the already-acquired difference information between the HF and LF models during the metamodelling process should be drawn more attention to, especially when the computational cost is limited. Previous sequential sampling approaches in single-fidelity metamodelling can be classified into two types: (a) the sequential sampling approaches are proposed for special metamodels; for example, Lin et al. (2004) proposed a sequential exploratory experiment design method for Kriging metamodel; Wei, Wu, and Chen (2012) proposed a curvature-based sequential sampling

approach for RBF metamodel; Eason and Cremaschi (2014) proposed a mixed adaptive sampling algorithm for artificial neural networks; and (b) the sequential sampling approaches are general sequential sampling approaches for single-fidelity metamodel; for example, Crombecq and Dhaene (2010) combined genetic algorithms and Monte Carlo method for sequential sampling; Crombecq et al. (2011; Crombecq, Laermans, and Dhaene 2011) proposed a novel hybrid sequential strategy (LOLA-Voronoi) for sequential sampling; Xu et al. (2014) presented a robust error-pursuing sequential sampling approach for global metamodelling; Yang and Xue (2015) proposed a weighted sequential sampling for global optimisation. So far, there is no efficient sequential sampling approach for VF metamodelling.

In this paper, an active learning global VF metamodelling approach based on ensemble of metamodels (EMs) and objective-oriented sequential sampling (OSS-EMs-VF) is proposed. In the proposed OSS-EMs-VF metamodelling approach, a model fusion technique based on EMs is employed to construct a VF metamodel to integrate the information from both LF and HF models. Furthermore, an active learning strategy is introduced in the metamodelling process to exploit the already-acquired information from the previous VF data to guide the VF metamodelling. The already-acquired information represents the location of regions where the differences between the HF and LF models are multi-model, non-smooth and have abrupt changes. The approximation performance of OSS-EMs-VF approach is demonstrated using some mathematical and engineering cases, and a rough comparison of OSS-EMs-VF approach and other VF metamodelling techniques for accuracy and robustness performances are made. Different problem types and influence of the sample sizes on metamodel performance are also explored. The main advantages of OSS-EMs-VF for VF metamodelling applications are analysed and summarised.

The rest of this paper is organised as follows. In Section 2, some background and definitions are provided. Details of the proposed OSS-EMs-VF metamodelling approach are described in Section 3. Numerical test cases section containing data sampling, performance measures, and the corresponding results and comparisons are provided in Section 4. As an illustration, an engineering case is studied in Section 5, followed by concluding remarks and future work in Section 6.

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2. Background and definition

In this section, some background and definitions used throughout this paper are presented.

2.1. Stand-alone metamodelling methods

Generally, the relationship between a vector of input variables $\boldsymbol{x}$ and corresponding output values Y can be expressed as:

$$Y = \hat{f}(\boldsymbol{x}, \beta) + \varepsilon, \tag{1}$$

where $\hat{f}(\cdot)$ is the approximation model, β represents the vector of coefficients, and ε denotes a stochastic factor. Metamodelling technologies differ with each other as to their choices of approximation models and stochastic process expressions. In this paper, three popular metamodelling methods, including Kriging, RBF and SVR, are adopted.

Detailed introduction of them can refer to the literature, for example, Acar (2015) and Zhou, Ma, and Li (2011)

2.2. VF metamodelling

The VF metamodelling technology is based on the assumption that, apart from a HF model that is sufficiently accurate but requires expensive computational cost, there is another one that is less accurate but is considerably less computationally demanding (Viana et al. 2014). A LF can be obtained by several ways (Yamazaki and Mavriplis 2012; Zheng et al. 2013; Mullins and Mahadevan 2014): (a) simplifying the analysis model (e.g. by using a coarse finite element mesh instead of a refined finite element mesh); (b) simplifying the modelling concept (e.g. by using a two-dimensional (2D) model instead of a three-dimensional (3D) model); and (c) simplifying the mathematical or physical description (e.g. by using the Euler non-cohesive equations instead of the Navier-Stokes viscous Newton equations). In the VF modelling process, a LF model is calibrated using the HF model responses from a suitable size of design experiments. In this way, the VF metamodels can make use of the advantages of both LF and HF models, that is, LF models are used to reduce the computational cost, while HF models are used to guarantee the accuracy.

Generally, the VF metamodel based on the interaction of HF model and LF model can be expressed as follows:

$$\hat{F}(\boldsymbol{x}, \boldsymbol{a}) \equiv \hat{F}(f^{l}(\boldsymbol{x}), \boldsymbol{a}) \approx F(\boldsymbol{x}), \tag{2}$$

where $\boldsymbol{x}$ is the design vector, $\hat{F}(\boldsymbol{x}, \boldsymbol{a})$ denotes the VF metamodel that is used to replace the actual HF model, $f^{l}(\boldsymbol{x})$ represents the response of the LF model and $F(\boldsymbol{x})$ represents the true response of the HF metamodel, $\boldsymbol{a}$ is a vector of tuning parameters used for minimising the discrepancy between the LF and HF models. From the above definition, the VF metamodel $\hat{F}(\boldsymbol{x}, \boldsymbol{a})$ tends to approach the high accuracy of the HF model but at a considerably less computational effort.

Three types of the LF model scaling have been summarised by Zadeh, Toropov, and Wood (2009):

Type 1: Linear and multiplicative metamodel with two tuning parameters:

$$\text{Linear:} \quad \hat{F}(\boldsymbol{x}, \boldsymbol{a}) = a_0 + a_1 f^{l}(\boldsymbol{x}), \tag{3}$$

$$\text{Multiplicative:} \quad \hat{F}(\boldsymbol{x}, \boldsymbol{a}) = a_0 f^{l}(\boldsymbol{x})^{a_1}, \tag{4}$$

where tuning parameter $\boldsymbol{a} = (a_0, a_1)^{\mathrm{T}}$.

Type 2: Scaling functions:

$$\text{Multiplicative form:} \quad \hat{F}(\boldsymbol{x}, \boldsymbol{a}) = f^{l}(\boldsymbol{x}) C(\boldsymbol{x}, \boldsymbol{a}), \tag{5}$$

$$\text{Linearform:} \quad \hat{F}(\boldsymbol{x}, \boldsymbol{a}) = f^{l}(\boldsymbol{x}) + C(\boldsymbol{x}, \boldsymbol{a}), \tag{6}$$

where $C(\boldsymbol{x}, \boldsymbol{a})$ can be a low order polynomial in $\boldsymbol{x}$ with a vector of tuning parameters $\boldsymbol{a}$.

Type 3: Use of model inputs as tuning parameters:

$$\hat{F}(\boldsymbol{x}, \boldsymbol{a}) = f(\boldsymbol{x}, \boldsymbol{a}), \tag{7}$$

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where $\boldsymbol{a}$ can be obtained as follows:

$$\min \; G(\boldsymbol{a}) = \sum_{p=1}^{P} (w_p(f^h - \hat{F}(\boldsymbol{x}_p, \boldsymbol{a}))^2, \tag{8}$$

where w_p denotes the relative contribution of the information at sample points $\boldsymbol{x}_p$ and $f^h(\boldsymbol{x}_p)$ represents the response of the HF model.

Generally, type 1 and type 2 can be used in local VF modelling approaches, and type 2 can also be used globally by adopting global metamodels to approximate the scaling function $C(\boldsymbol{x}, \boldsymbol{a})$. These two types of VF metamodelling have been successfully applied to the field of design optimisation, and type 3 is based on a deeper understanding of a process being modelled, which can be useful but is problem dependent. More details can refer to the literature, for example, Viana et al. (2014) and Zadeh, Toropov, and Wood (2009).

2.3. Difference unstable region

A difference unstable region (DUR) is defined as a region where the differences between the HF and LF models are multi-model, non-smooth and/or have abrupt changes. A HF model is one that is able to accurately describe the physical features of the system but with an unaffordable computational expense. A LF model is capable of reflecting the most prominent characteristics of the system at a considerably less computationally demanding. As a result, discrepancies occur in some regions where LF model fails to accurately describe the system. An example of DURs is illustrated in Figure 1. In Figure 1, the black line represents the actual HF model that is needed to be fitted, while the blue dashed line represents the LF one. The discrepancies between the HF model and LF model are represented by the red curve. It can be observed that the corresponding response values (inside the ellipse) in the $x \in (0.18, 0.30)$ and $x \in (0.62, 0.77)$ change dramatically, so that the two regions correspond to two DURs. And these DURs contain one or more local

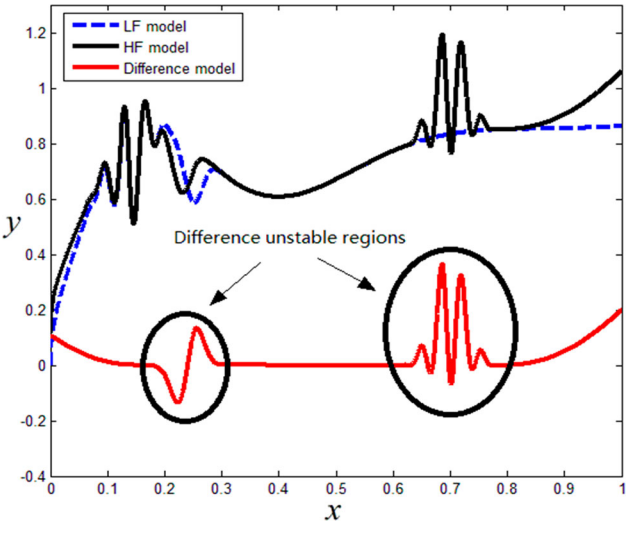


Figure 1. An example of difference unstable regions.

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peaks and troughs. In this regard, more points should be sampled in DURs for the HF model, which is used to tune the LF model, and relatively less attention should be paid to the regions where the discrepancies between the HF model and LF model are very little.

3. Proposed approach

3.1. Model fusion using EMs

One of the key issues for global VF metamodelling is how to combine the data stemming from both LF and HF models or how to use scale functions to calibrate the LF model according to a small amount of HF simulations (Han and Görtz 2012). Suppose that the HF sampling set is $\boldsymbol{x}^h = \{\boldsymbol{x}_1^h, \boldsymbol{x}_2^h, \ldots, \boldsymbol{x}_{m_h}^h\}$, which consists of m_h evaluated experiments in a K-dimensional design space, and its corresponding HF response is $\boldsymbol{f}^h = \{f_1^h, f_2^h, \ldots, f_{m_h}^h\}$. The scaling function value $\boldsymbol{c}(\boldsymbol{x})$ between the HF and LF model/metamodel for a HF sample point x_i^h can be calculated as:

$$c(\boldsymbol{x}_i^h) = f^h(\boldsymbol{x}_i^h) - f^l(\boldsymbol{x}_i^h), \quad i = 1, 2, \ldots m, \tag{9}$$

where $f^h(\boldsymbol{x}_i^h)$ is the actual response of the HF model for $\boldsymbol{x}_i^h$, $f^l(\boldsymbol{x}_i^h)$ can be either the original LF model or its metamodel. When a LF model is cheap to run, it can be directly used for VF modelling without fitting a metamodel to replace it. While at either of these circumstances: (a) running LF simulation is expensive; and (b) the relationships between the input variables and corresponding output values cannot be expressed explicitly in LF model, a LF metamodel is constructed instead of an actual LF model.

Since the expression or internal structure for the HF sampling set $\boldsymbol{x}^h$ and scaling function value $\boldsymbol{c}(\boldsymbol{x})$ is unknown, different stand-alone metamodels are used to fit the scaling functions; for example, Zheng et al. (2013) constructed a common Kriging metamodel for the scaling function. Tyan, Nguyen, and Lee (2015) adopted RBF network as the scaling function. Sun, Li, and Li (2012) put forward a SVR-based scaling function to accomplish the difference mapping. Main drawbacks of these practices are that the nature of the scaling function is not a priori and the accuracies of the constructed stand-alone metamodel largely depend on the current training sampling set and/or the problem properties at hand, so it cannot be guaranteed that the selected metamodel will always perform the best as another training data available or problems are changed. Therefore, as an alternative to using a stand-alone metamodel to fit the scaling function, it would be beneficial to replace the scaling function with EMs, which can take advantage of the prediction ability of each separate stand-alone metamodel. In this paper, three different stand-alone metamodelling techniques are considered here, and they include: Kriging, RBF, and SVR. The use of three candidate metamodels presents a good balance in performance and complexity. Two meta-models are not enough to provide enough variance, when one of the two metamodels is not suitable for the optimisation problem, leading to only one model and limiting the room for search efficiency improvement. On the other hand, four or more meta-models considerably increase the amount of computation, leading to poor efficiency. Each of the three selected metamodels, Kriging, RBF and SVR, presents a unique capability to cover a specific type of problems. Kriging is better for non-linear problems; the average accuracy and robustness of RBF are influenced only slightly by sample size, and SVR is more accurate for high-order problems.

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The EMs can be constructed by using a weighted average of different metamodels; hence, the metamodel for scaling function is defined as:

$$\hat{c}(\boldsymbol{x}) = \sum_{j=1}^{3} w_j \hat{c}_j(\boldsymbol{x}), \tag{10}$$

where $\hat{c}(\boldsymbol{x})$ is the predicted scaling function obtained from the ensemble metamodel, $\hat{c}_j(\boldsymbol{x})(j = 1, 2, 3)$ are the predicted scaling functions obtained using the Kriging, RBF and SVR metamodels, respectively.

For the Kriging metamodel, the Gaussian correlation function $\boldsymbol{R}(\cdot)$ is used for metamodelling. Once the vector of parameter Θ obtained by solving the following equation:

$$\begin{aligned} &\max\ \Phi(\Theta) = -\frac{[N \ln(\hat{\sigma}^2) + \ln|\boldsymbol{R}|]}{2}, \\ &\text{s.t. } \Theta > 0. \end{aligned} \tag{11}$$

$\hat{c}_1(\boldsymbol{x})$ can be obtained after several deductions:

$$\begin{aligned} \hat{c}_1(\boldsymbol{x}) &= \hat{\beta}_h + (\boldsymbol{r}_h)^{\mathrm{T}}(\boldsymbol{R}_h)^{-1}(\boldsymbol{f}^h(\boldsymbol{x}) - \boldsymbol{f}^l(\boldsymbol{x}) - \widehat{\beta}_h \boldsymbol{p}), \\ \hat{\beta}_h &= (\boldsymbol{p}^{\mathrm{T}}\boldsymbol{R}_h^{-1}\boldsymbol{p})^{-1}\boldsymbol{p}^{\mathrm{T}}\boldsymbol{R}_h^{-1}(\boldsymbol{f}^h(\boldsymbol{x}) - \boldsymbol{f}^l(\boldsymbol{x})). \end{aligned} \tag{12}$$

For the RBF metamodel, the optimal weight coefficients λ_i^* can be obtained by:

$$\lambda_i^* = (\Phi^{\mathrm{T}}\Phi + \Lambda)^{-1}\Phi^{\mathrm{T}}(f^h(\boldsymbol{x}) - f^l(\boldsymbol{x})). \tag{13}$$

In this study, popular Gaussian function is selected as the RBF $\phi(\cdot)$. With the optimal weight coefficients of the liner combinations λ_i^* obtained by the following equation:

$$\lambda_i = (\Phi^{\mathrm{T}}\Phi + \Lambda)^{-1}\Phi^{\mathrm{T}} y_i, \tag{14}$$

where Λ are all zero except for the regularisation parameters along its diagonal. Φ is the design matrix can be expressed as:

$$\Phi = \begin{pmatrix} \phi(\|x_1 - x_1\|) & \phi(\|x_1 - x_2\|) & \cdots & \phi(\|x_1 - x_m\|) \\ \phi(\|x_2 - x_1\|) & \phi(\|x_2 - x_2\|) & \cdots & \phi(\|x_2 - x_m\|) \\ \vdots & \vdots & \ddots & \vdots \\ \phi(\|x_m - x_1\|) & \phi(\|x_m - x_2\|) & \cdots & \phi(\|x_m - x_m\|) \end{pmatrix}. \tag{15}$$

Then $\hat{c}_2(\boldsymbol{x})$ can be obtained by the following expression:

$$\hat{c}_2(\boldsymbol{x}) = \sum_{i=1}^{m} \lambda_i^* \phi(\|\boldsymbol{x} - \boldsymbol{x}_i\|). \tag{16}$$

To construct SVR metamodel $\hat{c}_3(\boldsymbol{x})$, the Gaussian kernel function is adopted for its advantages of fewer parameters to be set and an excellent overall performance. The

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following quadratic programming should be solved to obtain the vector $\boldsymbol{\delta}$ and $\boldsymbol{\delta}^*$:

$$\begin{aligned} \max \quad & -\frac{1}{2}\sum_{i,j=1}^{l}\left(\delta_i-\delta_i^*\right)\left(\delta_j-\delta_j^*\right)\exp\left(\frac{\left\|\boldsymbol{x}_i-\boldsymbol{x}_j\right\|}{2\sigma^2}\right) \\ & -\varepsilon\sum_{i=1}^{l}\left(\delta_i+\delta_i^*\right)+\sum_{i=1}^{l}\left(f^h(\boldsymbol{x}_i)-f^l(\boldsymbol{x}_i)\right)\left(\delta_i-\delta_i^*\right), \\ \text{s.t.} \quad & \sum_{i=1}^{m}\left(\delta_i-\delta_i^*\right)=0, \\ & \left(\delta_i-\delta_i^*\right)\in[0,C], \\ & i,j=1,2,\ \ldots,m. \end{aligned} \tag{17}$$

When the optimal $\boldsymbol{\delta}$ and $\boldsymbol{\delta}^*$ are obtained, $\hat{c}_3(\boldsymbol{x}_i)$ can be obtained through:

$$\hat{c}_3(\boldsymbol{x})=\sum_{i=1}^{m}\left(\delta_i-\delta_i^*\right)\exp\left(\frac{\left\|\boldsymbol{x}_i-\boldsymbol{x}_j\right\|}{2\sigma^2}\right)+b\,;\quad i,j=1,2,\ \ldots,m. \tag{18}$$

In Equation (10), $w_j (j = 1, 2, 3)$ are weight factors that determine the relative contribution of the three metamodels in ensemble metamodel. To obtain unbiased response estimations, a constraint is posed on weight factors as follows:

$$\sum_{j=1}^{3} w_j = 1. \tag{19}$$

The weight factors can be determined such that the metamodel with high accuracy have large weight factors and vice versa (Acar 2015; Zhou, Ma, and Li 2011). In this paper, the weight factors are determined by minimising the generalised mean square leave-one-out (LOO) errors ($\mathrm{GMSE_{LOO}}$) of the ensemble metamodels. LOO method is one of the cross validation methods that can be used for assessing the accuracy of a metamodel. The basic processes of obtaining the $\mathrm{GMSE_{LOO}}$ can be divided into four steps.

Step 1: For a given HF sampling set $\boldsymbol{x}^h = \{\boldsymbol{x}_1^h, \boldsymbol{x}_2^h, \ldots, \boldsymbol{x}_{m_h}^h\}$, remove one of the sampling points out.

Step 2: Use the remaining sampling points $\boldsymbol{x}_{-i}^h = \{\boldsymbol{x}_1^h, \boldsymbol{x}_2^h, \ldots, \boldsymbol{x}_{i-1}^h, \boldsymbol{x}_{i+1}^h, \ldots, \boldsymbol{x}_{m_h}^h\}$ to build the EMs, which is used to predict the response value for the omitted sample.

Step 3: Calculate the square difference between the predicted values and the accurate value for the omitted sample.

Step 4: Repeat Step 1 to Step 3 until all sample points in the given sampling set are considered.

With this method, no additional simulation runs are needed for error calculations. The $\mathrm{GMSE_{LOO}}$ for the EMs can be expressed as:

$$\mathrm{GMSE_{LOO}} = \sum_{i=1}^{m}\frac{\left(\hat{c}_{-i}(\boldsymbol{x}_i)-c(\boldsymbol{x}_i)\right)^2}{m}, \tag{20}$$

where $c(\boldsymbol{x}_i)$ denotes the actual response value at x_i, $\hat{c}_{-i}(x_i)$ denotes the prediction of the

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response value for x_i using the EMs created based on the current sampling set with the sample point x_i moved out.

Combining the weighted-sum formulations of the ensemble metamodels in Equation (10) and error prediction metric in Equation (20), a constrained minimisation problem is established to select the optimum weight factors. It can be expressed as follows:

$$\begin{aligned} &\text{find} \quad w_j, \\ &\min \quad \mathrm{GMSE}_{\mathrm{LOO}}(w_j) = \sum_{i=1}^{m} \left(\sum_{j=1}^{3} w_j \hat{c}_j(\boldsymbol{x}_i^h) - c(\boldsymbol{x}_i^h) \right)^2 / m, \\ &\text{s.t.} \quad \sum_{i=1}^{3} w_j = 1. \end{aligned} \tag{21}$$

Since the optimisation problem in Equation (21) is not necessarily convex, evolutionary algorithms are preferred to solve this problem. In this study, genetic algorithms (Coello Coello 2000) and a penalty method (Homaifar, Qi, and Lai 1994) for constraint handling are used.

With the obtained optimal weight factors, the formulation of the proposed VF metamodelling approach becomes:

$$\hat{y}_v(\boldsymbol{x}) = y^l(\boldsymbol{x}) + c(\boldsymbol{x}) = y_l(\boldsymbol{x}) + \sum_{j=1}^{3} w_j^* \hat{c}_j(\boldsymbol{x}). \tag{22}$$

3.2. Active learning strategy

The proposed active learning strategy is based on the observation that the more local extreme points taken as the sample points for scaling function, the more accurate a VF model is. As described in Section 2.3 these local extreme points are assumed to be located in the DURs. However, the HF models and/or LF models are usually black-box functions; it is infeasible to intuitively observe the locations of DURs in which only a small amount of sample points have been sampled. Hence, a sequential single objective formulation is constructed in the active learning process to probe the DURs and adaptively select new sample points. In this formulation, the optimisation objective is to select a sample point with the maximum predicted error obtained by analysing data (metamodels and sample points) from previous iterations, and a space-filling criterion is introduced and treated as a constraint to avoid generating clustered sample points. It is to be noted that the proposed active learning strategy introduced in OSS-EMs-VF is general in the sense that it can be integrated with any VF scaling metamodelling technique.

In this subsection, we present the proposed active learning process with several parts. They are: (1) error prediction metrics for estimated points, (2) criterion for space filling, and (3) selection of the next new sample points.

3.2.1. Error prediction metrics for estimated points

To avoid extra computational cost, the previously obtained LOO error for the sample points in the current sampling set will be exploited to assess the accuracy of a metamodel. Clearly, the LOO method is not available for directly estimating the prediction error for

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those sample points that are not in the current sampling set. However, it can be used to measure the sensitivity of the metamodel to the lost information (Li, Aute, and Azarm 2010), that is, loss of $\boldsymbol{x}_i^h$ in the current sampling set. In this regard, the LOO error for an existing sample point $\boldsymbol{x}_i^h$ can allow us to identify DURs where more sample points need to be added to improve accuracy of the VF metamodel. It is observed that an estimated point has a more similar performance (e.g. LOO error) with an existing sample point, which is closer to it than others. Thus, the LOO error for an estimated point will be likely to be large if it is close to those existing sample points, which are located in a DUR sampled by a small amount of points. Based on this observation, we propose an error prediction metric termed as weighted cumulative error. The basic idea of this metric is that the predicted error for an estimated point can be expressed using the weighted cumulative error for the sample points in the current set. The closer the distance between the estimated point and the existing sampling point, the larger the weighting factor. For a given HF sampling set $\boldsymbol{x}^h = \{\boldsymbol{x}_1^h, \boldsymbol{x}_2^h, \ldots, \boldsymbol{x}_{m_h}^h\}$, the weighted cumulative error ($\delta(\boldsymbol{x}_o^h)$) on estimated point $\boldsymbol{x}_o^h$ is estimated by:

$$\delta(\boldsymbol{x}_o^h) = \sum_{i=1}^{n} w_i \left| \hat{c}_{-i}(\boldsymbol{x}_o^h) - \hat{c}(\boldsymbol{x}_o^h) \right|. \tag{23}$$

where $\hat{c}(\boldsymbol{x}_o^h)$ denotes the predicted response value for $\boldsymbol{x}_o^h$ using the EMs created based on all existing sample points. $\hat{c}_{-i}(\boldsymbol{x}_o^h)$ denotes the prediction of the response value for $\boldsymbol{x}_o^h$ using the EMs created based on the current sampling set with the sample point $\boldsymbol{x}_i^h$ moved out. w_i represents the weighting factor reflecting the distance between the estimated point $\boldsymbol{x}_o^h$ and the sample point $\boldsymbol{x}_i^h$. w_i is calculated by:

$$w_i = \frac{a^{-d(\boldsymbol{x}_i^h, \boldsymbol{x}_o^h)}}{a^{-\sum_{i=1}^{m_h} d(\boldsymbol{x}_i^h, \boldsymbol{x}_o^h)}}, \tag{24}$$

where $d(\boldsymbol{x}_i^h, \boldsymbol{x}_o^h)$ is defined to be the Euclidean distance. a is a constant coefficient that is greater than 1; in this paper, a is set to be the natural base, i.e. $a = e$.

3.2.2. Criterion for space filling

It is to be noticed that the sample points may cluster if selected just according to the prediction error using Equation (23). This is because of the mode we choose the new sampling point: The closer $\boldsymbol{x}_o^h$ is to some existing sample points, the larger weighted cumulative error. As a consequence, newly added points will firstly cluster in one DUR area and then consider other DURs. Given the limited resources, some DURs may not have a chance to be sampled. To avoid this clustering problem, a space-filling criterion is introduced. The space-filling criterion described by the Euclidean distance in the design space can be obtained in a three-step procedure. The steps are as follows.

Step 1: For each existing sample point, calculate the minimum distance of this point from the remaining sample points in the current sampling set $\boldsymbol{x}^h$.
Step 2: Calculate the average of the minimum distance.
Step 3: Set the space-filling criterion to be:

$$\left\| \boldsymbol{x}_o^h - \boldsymbol{x}_i^h \right\|_2 \geq \beta \cdot \text{mean}\left(min(\left\| \boldsymbol{x}_i^h - \boldsymbol{x}_j^h \right\|_2)\right), \forall \boldsymbol{x}_i^h, \boldsymbol{x}_j^h \in \boldsymbol{x}^h \cap (i \neq j), \tag{25}$$

where β is the space-filling factor. Generally, too large β forces the added sample points to distribute evenly, but does not contribute to focusing on the DURs. Too small β fails to prevent the added sample points from clustering. It has been found by performing a large number of mathematical investigation that β ($0.2 \leq \beta \leq 0.5$) is proper in most cases, and we adopt ($\beta = 0.3$) in all test cases presented in this paper.

3.2.3. Selection of the next new sample points

The next new sample points can be generated with a generalised objective-oriented optimisation problem, which combines the error prediction metric and the space-filling criterion in the previous two steps. The generalised objective-oriented optimisation problem can be expressed as follows:

$$\begin{aligned} &\max \quad \delta(\boldsymbol{x}_o^h) = \sum_{i=1}^{n} w_i \left| \hat{c}_{-i}(\boldsymbol{x}_o^h) - \hat{c}(\boldsymbol{x}_o^h) \right|, \\ &\text{s.t.} \quad \left\| \boldsymbol{x}_o^h - \boldsymbol{x}_i^h \right\|_2 \geq \beta \cdot \text{mean}\left(min(\left\| \boldsymbol{x}_i^h - \boldsymbol{x}_j^h \right\|_2) \right), \forall \boldsymbol{x}_i^h, \boldsymbol{x}_j^h \in \boldsymbol{x}^h \cap (i \neq j). \end{aligned} \tag{26}$$

The expression above can guarantee a reasonable distance between the new sample points and the existing sampling points, thus improving the accuracy of the VF metamodel. There may exist multiple local minima for the optimisation problem in Equation (26), thus genetic algorithms (Coello Coello 2000) and a penalty method (Homaifar, Qi, and Lai 1994) for constraint handling are used to solve this problem.

3.3. Steps for the proposed OSS-EMs-VF metamodelling

The flowchart of the proposed OSS-EMs-VF is demonstrated in Figure 2. The detailed steps are as follows:

Step 1: Generate two sampling sets $\boldsymbol{x}^l$ and $\boldsymbol{x}^h$. One set ($\boldsymbol{x}^l$) with a large number of sample points is used to obtain response values for LF model, the other one ($\boldsymbol{x}^h$) with a significantly smaller set is used to evaluate the HF model.
Step 2: Run LF simulation model at the $\boldsymbol{x}^l$ to obtain the LF response values.
Step 3: Run HF simulation model at the $\boldsymbol{x}^h$ to obtain the HF response values.
Step 4: Build metamodel for the LF model based on the sampled LF data.
Step 5: Construct the scaling function using EMs. This step can be divided into several steps to demonstrate the approach in more details. The details of each step are given next and the entire procedures are demonstrated in the dashed box of Figure 2.
Step 5.1: Calculate the discrepancies between the HF model and tuned LF metamodel for sample points using Equation (9).
Step 5.2: Build three stand-alone metamodels (Kriging, RBF, SVR) for the scaling function.
Step 5.3: Minimise the generalised mean square LOO errors to obtain the optimum weight factors by solving the optimisation problem in Equation (21).
Step 6: Build the VF metamodel using Equation (22).

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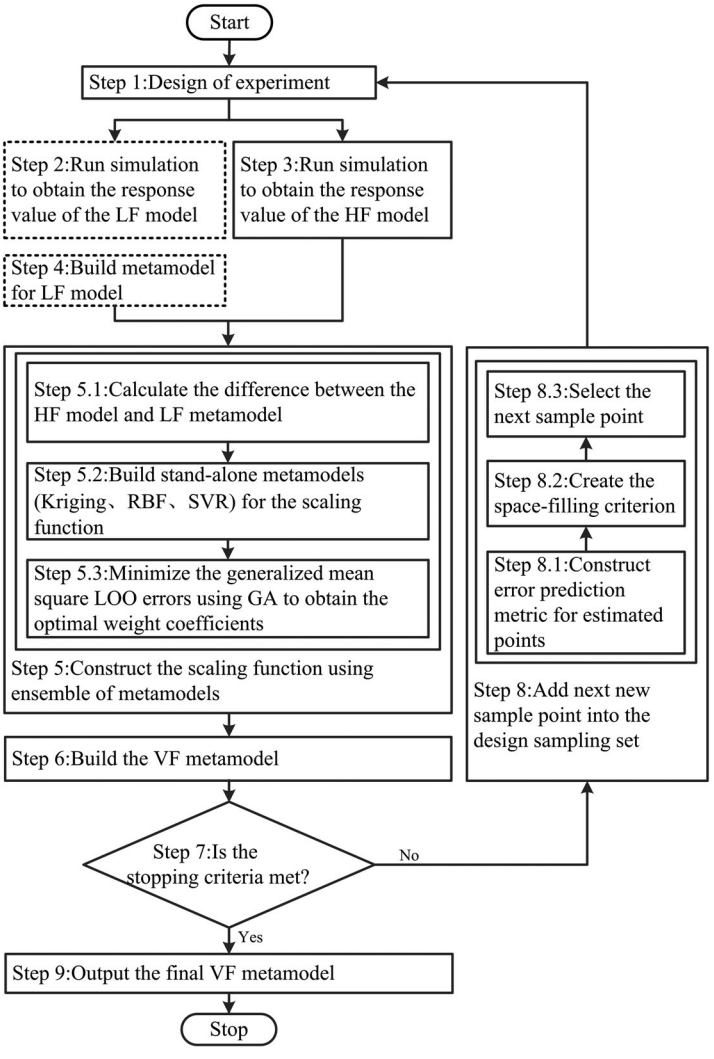


Figure 2. Flow chart for the proposed approach.

Step 7: Check the stopping criterion. Check whether the pre-set maximum number of sample points is achieved: If yes, go to Step 9; otherwise, go to Step 8.
Step 8: Add next new sample point in the sampling set ($\boldsymbol{x}^h$) by solving Equation (26).
Step 9: Output the final VF metamodel.

4. Numerical examples and results

4.1. Data sampling

The goal of the data sampling is to minimise the influence of errors in physical experiments on the response functions, while allowing the designers to build metamodels more efficiently (Zadeh, Toropov, and Wood 2009). In this work, the Optimal Latin

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Hypercube Sampling (OLHS), proposed by Jin et al. (2005), is adopted to generate sample points. Two different sample sizes, small size ($m = 20$) and large size ($m = 60$), are considered. It should be noticed that the defined large sample size for HF model is still in the scope of small size compared with that for LF model. For the proposed OSS-EMs-VF metamodelling approach, a basic requirement for the initial sampling set is that the design space should be uniformly covered so that all DURs have an equal chance to be discovered. In this work, half of the numbers of pre-set total sample points are generated as an initial HF sampling set to get a full knowledge of the discrepancy between HF and LF models.

4.2. Accuracy and robustness assessment

Two accuracy metrics, relative root mean square error (RRMSE) and relative maximum absolute error (RMAE), are adopted to measure the accuracy of each metamodel. RRMSE and RMAE represent the global accuracy and local accuracy of the metamodel, respectively. The lower the value of RRMSE/RMAE, the higher accuracy of the metamodel. Expressions of these two accuracy metrics are as follows:

$$\begin{aligned} \mathrm{RMAE} &= \frac{1}{\mathrm{STD}} \max\left|y_i - \hat{y}_i\right|, i = 1, \ldots, N, \\ \mathrm{RRMSE} &= \frac{1}{\mathrm{STD}} \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}, \\ \mathrm{STD} &= \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(y_i - \bar{y}_i)^2}, \end{aligned} \tag{27}$$

where N represents the total number of test points; y_i is the actual response value at test points; $\hat{y}_i$ is the predicted response at test points; $\bar{y}_i$ and STD are the mean and standard deviation of all observed response values, respectively.

4.3. Numerical experiments

Eight well-known numerical test problems from Zheng et al. (2013), Dong et al. (2014) and Zhou et al. (2015a, 2015b) are used to demonstrate the efficiency of the proposed metamodelling approach. These numerical test problems are classified into two types. The first type is a set of test examples, whose difference response features between HF and LF are flat; the second type is a set of test cases with multiple DURs. The eight test cases are summarised in Table 1 and their mathematical functions are described below. In all tests, y_h represents the HF metamodel, which is needed to be approximated and y_l represents the LF one. The LF models of each numerical example are directly used for VF modelling without fitting a metamodel to replace it.

Table 1. Summary of the additional mathematical test cases.

Types	Test functions	Dimensions	Difference characteristics
Type 1	Six-hump Camel-back function (SC)	2	Difference model between HF and LF is flat
	Modified Branin (MBR)	2	
	2D Rosenbrock	2	
	Dixon & Price function (DP)	4	
Type 2	Rastrigin's function	2	Difference model between HF and LF is flat in general and with some DURs
	Sinusoidal function	2	
	Hartman function (Hart6)	6	
	Borehole model function	8	

- Six-hump Camel-back function (SC):

$$\begin{aligned} f(x_1, x_2) &= 4x_1^2 - 2.1x_1^4 + x_1^6/3 + x_1x_2 - 4x_2^2 + 4x_2^4; \\ y_h &= f(x_1, x_2); \\ y_l &= f(0.7x_1, 0.7x_1) + x_1x_2 + \sin(x_1) + 2x_2 + \sin(x_2) + x_1; \\ x_1 &\in [-2,\ 2], x_2 \in [-2,\ 2]. \end{aligned} \tag{28}$$

- Modified Branin (MBR):

$$\begin{aligned} f(x_1, x_2) &= 10 + [x_2 - 5.1 \times x_1^2/(4\pi^2) + 5x_1/\pi - 6]^2 + 10\cos(x_1)[1 - 1/8\pi]; \\ y_h &= f(x_1, x_2) - 22.5x_2; \\ y_l &= f(0.7x_1, 0.7x_2) - 15.75x_2 + 20(0.9 + x_1)^2 - 50; \\ x_1 &\in [-5, 10], x_2 \in [0, 15]. \end{aligned} \tag{29}$$

- 2-dimensional Rosenbrock function (2D Rosenbrock):

$$\begin{aligned} f(x_1, x_2) &= 100(x_1^2 - x_2)^2 + (x_1 - 1)^2; \\ y_h &= f(x_1, x_2); \\ y_l &= 70(x_1^2 - x_2)^2 + 0.7(x_1 - 1)^2; \\ x_1 &\in [-2.048, 2.048], x_2 \in [-2.048, 2.048]. \end{aligned} \tag{30}$$

- Dixon & Price function (DP):

$$\begin{aligned} f(x_1, \ldots, x_4) &= (x_1 - 1)^2 + \sum_{i=2}^{n} i(2x_i{}^2 - x_{i-1})^2; \\ y_h &= f(x_1, \ldots, x_4); \\ y_l &= (x_1 - 1)^2 + \sum_{i=2}^{n} i(2x_i{}^2 - 0.75x_{i-1})^2; \\ x_i &\in [-10,\ 10],\ \ i = 1, \ldots, 4; \\ n &= 4. \end{aligned} \tag{31}$$

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- Rastrigin's function:

$$\begin{aligned} f(x_1, x_2) &= 20 + x_1^2 + x_2^2 - 10(cos(2\pi x_1) + cos(2\pi x_2)); \\ y_h &= f(x_1, x_2); \\ A &= f(x_1, x_2) \cdot (sign(x_2) + 0.5)/abs(sign(x_2) + 0.5); \\ B &= 10 + x_1^2 - 10cos(2\pi x_1); \\ C &= B \cdot (sign(-x_2) + 0.5)/abs(sign(-x_2) + 0.5); \\ y_l &= \begin{cases} A & \text{when } A \geq C \\ C & \text{when } A < C' \end{cases} \\ x_1 &\in [-1, 1], x_2 \in [-1, 1]. \end{aligned} \tag{32}$$

- Sinusoidal function:

$$\begin{aligned} f(x_1, x_2) &= x_1 \sin(x_2) + x_2 \sin(x_1); \\ y_h &= f(x_1, x_2); \\ y_l &= 0.8x_1 \sin(0.9x_2) + 0.9x_2 \sin(0.8x_1); \\ x_1 &\in [-2\pi, 2\pi], x_2 \in [-2\pi, 2\pi]. \end{aligned} \tag{33}$$

- Hartman function with $n = 6$ (Hart6):

$$\begin{aligned} f(x_1, \ldots x_6) &= -\sum_{i=1}^{4} c_i \exp\left[-\sum_{j=1}^{6} a_{ij}(x_j - p_{ij})^2\right], x_j \in [0, 1]; \\ [c_i] &= [1 \quad 1.2 \quad 3 \quad 3.2]^{\mathrm{T}}, [a_{ij}] = \begin{bmatrix} 10 & 3 & 17 & 3.05 & 1.7 & 8 \\ 0.05 & 10 & 17 & 0.1 & 8 & 4 \\ 3 & 3.5 & 1.7 & 10 & 17 & 8 \\ 17 & 8 & 0.05 & 10 & 0.1 & 14 \end{bmatrix}; \\ [p_{ij}] &= \begin{bmatrix} 0.1312 & 0.1696 & 0.5569 & 0.0124 & 0.8283 & 0.5886 \\ 0.2329 & 0.4139 & 0.8307 & 0.3736 & 0.1004 & 0.9991 \\ 0.2348 & 0.1451 & 0.3522 & 0.2883 & 0.3047 & 0.6650 \\ 0.4047 & 0.8828 & 0.8732 & 0.5743 & 0.1091 & 0.0381 \end{bmatrix}; \\ y_h &= f(x_1, \ldots x_6); \\ y_l &= -\sum_{i=1}^{4} lc_i \exp[-\sum_{j=1}^{6} a_{ij}(l_j x_j - p_{ij})^2], x_j \in [0, 1]; \\ [lc_i] &= [1.1 \quad 0.8 \quad 2.5 \quad 3]^T, [l_j] = [0.75 \quad 1 \quad 0.8 \quad 1.3 \quad 0.7 \quad 1.1]^{\mathrm{T}}. \end{aligned} \tag{34}$$

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- Borehole model function:

$$
\begin{aligned}
f_{\text{borehole}} &= \frac{2\pi x_3(x_4 - x_6)}{\ln(x_2/x_1)[1 + 2x_7x_4/(\ln(x_2/x_1)x_1^2x_8) + x_3/x_5]};\\
y_h &= f_{\text{borehole}}(x);\\
y_l &= 0.4\, f_{\text{borehole}}(x) + 0.07x_1^2x_8 + x_1x_7/x_3 + x_1x_6/x_2 + x_1^2x_4;\\
x_1 &\in [0.05, 0.15],\ x_2 \in [100, 50000];\\
x_3 &\in [63070, 115600],\ x_4 \in [990, 1110];\\
x_5 &\in [63.1, 116],\ x_6 \in [700, 820];\\
x_7 &\in [1120, 1680],\ x_8 \in [9855, 12045].
\end{aligned}
\tag{35}
$$

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4.4. Results and comparisons

To compare the performance of each VF metamodelling technique under different circumstances (e.g. different sample sizes, problem types), additional 1600 randomly validation points are used in each numerical case. Based on these validation data, each accuracy metric is calculated to illustrate the average accuracy and robustness of each VF metamodelling technique, respectively. For comparison, we model all tests with three other VF metamodelling methods: (1) VF metamodelling approach based on Kriging scaling function (Kriging-VFM); (2) VF metamodelling approach based on RBF scaling function (RBF-VFM); and (3) VF metamodelling approach based on SVR scaling function (SVR-VFM). Besides, the proposed OSS-EMs-VF approach without using active learning strategy (EMs-VF) is also constructed for comparison. To avoid unrepresentative numerical results, 15 different runs are carried out for each example and the average accuracy metrics are selected as the final results. The accuracy results of different metamodelling approaches are summarised in Table 2. In Table 2, the best results for each model type are printed in bold, while the worst results for each model type are printed in bold-italic. Two intuitive conclusions can be drawn from Table 2, that is, (1) the VF metamodelling approach, whose scaling function based on EMs (OSS-EMs-VF and EMs-VF) is never the worst one among the five VF metamodelling approaches for all the numerical cases at two different sample sizes, while other three VF metamodel (Kriging-VF, RBF-VF and SVR-VF) are the worst at least for one case, which indicates that the proposed model fusion technique based on EMs can decrease the risk of adopting an inappropriate metamodel. (2) For most test cases, metamodels constructed by OSS-EMs-VF are demonstrated to be the best in terms of both local and global accuracy.

4.4.1. Overall performance

Figure 3 depicts the mean and standard deviation of the two accuracy metrics for all the metamodels, which consider different types of problems under small and large sample sizes. As seen in Figure 3(a), it can be observed that the accuracy of OSS-EMs-VF approach is the best in terms of both RMAE and RRMSE metrics. EMs-VF performs better than Kriging-VF, RBF-VF and SVR-VF. Kriging-VF approach, with the largest average RMAE value, exhibits a poorest local approximation performance. The average value of RRMSE

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Table 2. Summary of the accuracy results for different test functions.

		Small set ($n = 20$)					Large set ($n = 60$)				
Test functions	Accuracy metrics	OSS-EMs-VF	EMs-VF	Kriging-VF	RBF-VF	SVR-VF	OSS-EMs-VF	EMs-VF	Kriging-VF	RBF-VF	SVR-VF
SC	RMAE	**0.693**	1.071	***2.128***	1.628	1.733	**0.143**	0.307	***1.336***	0.238	0.478
	RRMSE	0.204	**0.137**	0.307	0.251	***0.380***	**0.035**	0.036	***0.183***	0.020	0.060
MBR	RMAE	**0.228**	0.727	***0.849***	0.685	0.726	**0.097**	0.191	0.472	***0.610***	0.182
	RRMSE	**0.075**	0.084	***0.168***	0.093	0.088	**0.022**	0.041	***0.089***	0.058	0.045
2D Rosenbrock	RMAE	**0.299**	0.657	***1.114***	0.316	0.499	**0.027**	0.111	***0.608***	0.146	0.101
	RRMSE	**0.063**	0.074	***0.116***	0.081	0.089	**0.004**	0.011	***0.096***	0.012	0.013
DP	RMAE	**0.044**	0.059	0.061	***0.067***	0.049	**0.005**	0.033	0.051	***0.073***	0.052
	RRMSE	**0.009**	0.012	0.013	***0.014***	0.010	**0.004**	0.007	0.008	***0.010***	0.009
Rastrigin's	RMAE	**0.919**	1.097	1.170	1.343	***1.559***	**0.116**	0.381	0.156	***0.719***	0.223
	RRMSE	0.279	**0.279**	0.300	***0.420***	0.348	**0.014**	0.046	0.025	0.129	***0.048***
Sinusoidal	RMAE	**1.336**	2.166	***3.182***	2.715	2.164	**0.713**	2.193	2.091	2.289	***2.349***
	RRMSE	0.427	**0.427**	0.501	***0.665***	0.475	**0.228**	0.305	***0.384***	0.331	0.333
Hart6	RMAE	**1.944**	2.196	2.165	***2.688***	2.171	**1.849**	1.944	***2.131***	2.023	2.010
	RRMSE	**0.482**	0.528	***0.668***	0.646	0.544	0.481	**0.461**	***0.579***	0.574	0.511
Borehole	RMAE	**2.029**	2.144	***2.959***	2.057	2.173	**0.416**	1.567	***2.291***	2.226	1.522
	RRMSE	0.805	**0.789**	0.816	***0.872***	0.797	**0.109**	0.530	0.649	***0.792***	0.711

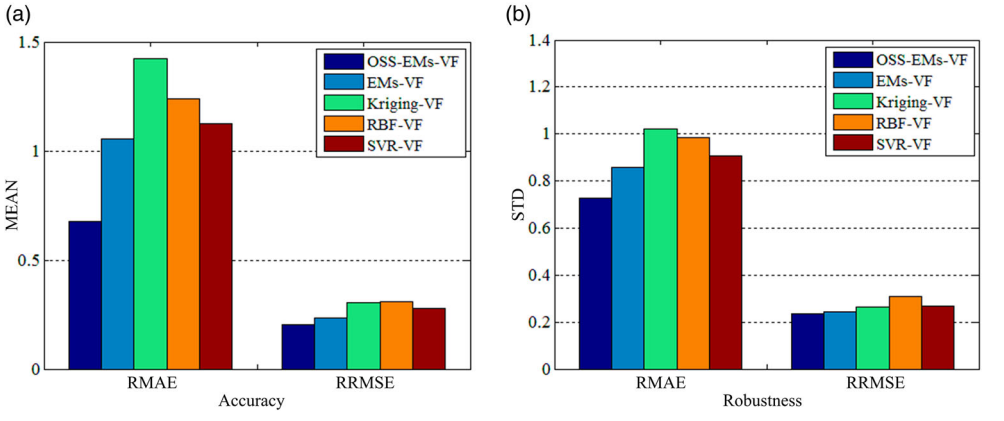


Figure 3. Overall performance for different VF metamodelling approaches: (a) accuracy, and (b) robustness.

is the largest for RBF-VF, which indicates RBF-VF has the poorest global performance among five VF metamodelling approaches. As illustrated in Figure 3(b), the robustness of the OSS-EMs-VF is also the best in the two accuracy metrics. EMs-VF performs the next best, followed by the SVR-VF. RBF-VF is slightly better than Kriging-VF in RMAE, while Kriging-VF is a little better than DMF-Kriging in RRMSE. In summary, the performance of OSS-EMs-VF is the best in terms of both average accuracy and robustness for all the numerical test cases at small and large sample sizes.

4.4.2. Performance for different types of problems under small and large sample sizes

Figures 4 and 5 demonstrate the accuracy and robustness of each metamodelling performance for different types of problems under different sample sizes. In Figures 4 and 5, 'Type 1' and 'Type 2' represent the types of the problems, while 'Large' and 'Small' represent sizes of the sample points. So for example, 'Type 1 Large' means the problem belongs to Type 1 with large sample sizes. As can be seen in Figure 4(a) and 4(b) that the local average accuracy and robustness of OSS-EMs-VF are significantly better than other four VF metamodelling methods, especially for Type 2 problems with large sample sizes. EMs-VF performs better than Kriging-VF, RBF-VF and SVR-VF. Kriging-VF shows the worst local performance for different types of problems under different sample sizes, except that its local average accuracy is slightly better than RBF-VF for Type 2 problems with large sample sizes. As shown in Figure 4(c) and 4(d), OSS-EMs-VF performs the best in terms of global accuracy and robustness except for its slightly lower global accuracy and robustness than EMs-VF for Type 1 problems with small sample sizes. This can be explained by the fact that the difference models of these test cases are smooth everywhere. As a result, they are in favour of metamodelling methods that focus on filling the entire input space uniformly. RBF-VF shows the worst global performance for Type 2 problems under different sample sizes.

In Figure 5, the accuracy and robustness are derived for single contributing factors, that is, problem types and different sample sizes. It indicates that all VF metamodelling approaches perform significantly better for Type 1 problems than Type 2 problems.

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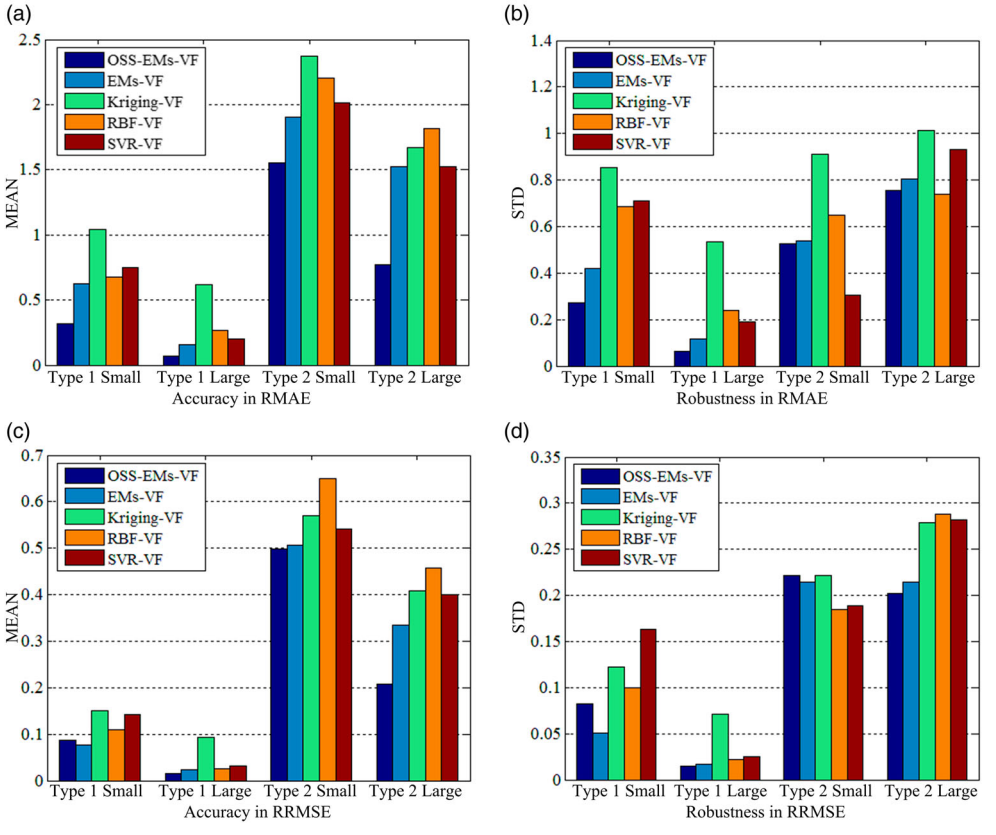


Figure 4. Performances for different types of problems under small and large sample sizes: (a) accuracy in RMAE; (b) robustness in RMAE; (c) accuracy in RRMSE; and (d) robustness in RRMSE.

Meanwhile, it can be easily found that when the sample set is increased from small size to large size, the average accuracy and robustness of all the VF metamodelling are increased. For small sample size, the local accuracy and robustness of OSS-EMs-VF are the best, while its global accuracy and robustness are close to those of EMs-VF. Kriging-VF shows the worst performance in local accuracy and robustness, while its global accuracy and robustness are slightly better than RBF-VF. For large sample size, it is observed that the proposed OSS-EMs-VF approach significantly outperforms other approaches in terms of both RMAE and RRMSE. EMs-VF achieves the next best approximation performance. Considering the problem types, the proposed OSS-EMs-VF metamodelling approach outperforms the other methods for Type 2, which is attributed to its active learning strategy that is able to adaptively allocate more HF sample points in the DURs. While the proposed OSS-EMs-VF approach has a comparative performance to EMs-VF for Type 1 whose difference response features between HF and LF are flat.

4.5. Sensitivity analysis of the number of initial sample points for OSS-EMs-VF

An important factor which may impact on the success of the proposed modelling method is the number of initial sample points. Some problems may prefer less initial sample points

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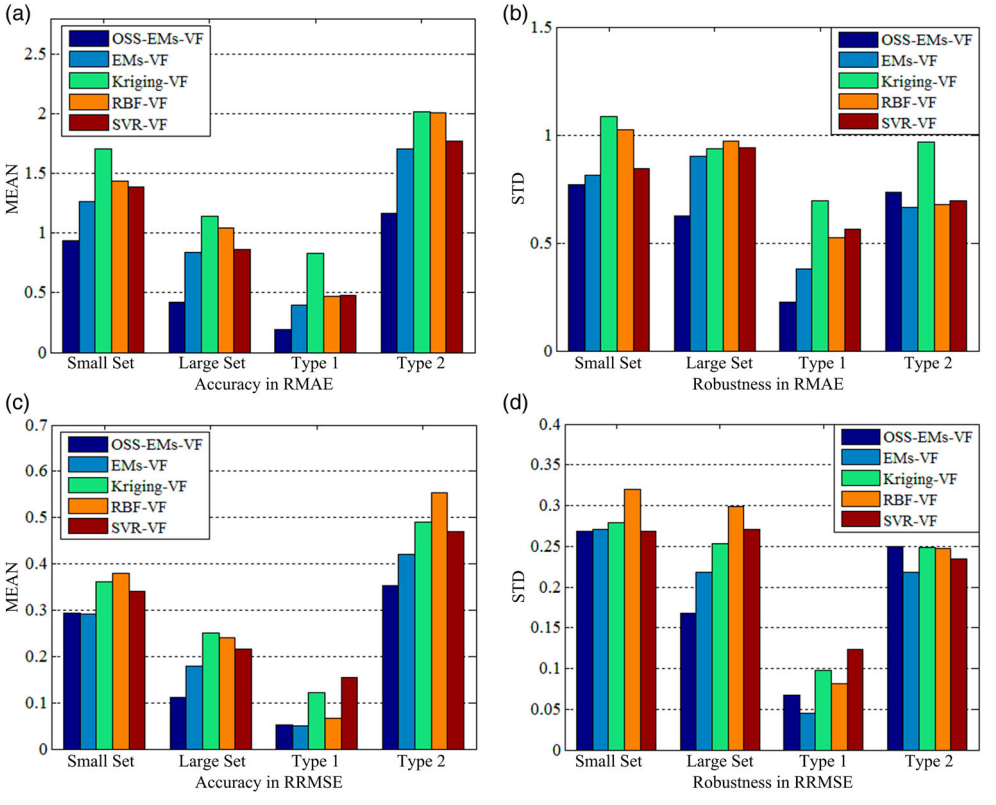


Figure 5. Performances comparison for single contributing factors: (a) accuracy in RMAE; (b) robustness in RMAE; (c) accuracy in RRMSE; and (d) robustness in RRMSE.

(allowing more sample points selected in the iterative process) while others may require more. It is suggested that the appropriate number of the initial sample points depends on the problem properties at hand and dimension of the problems.

To study the impacts of the number of initial sample points on the performance of the proposed adaptive VF metamodelling approach, five levels of initial sample sizes, 30%, 40%, 50%, 60% and 70% of the total number of sample points, are considered for the above eight test functions. Taking the average results of 50% of the total sample points as the baseline, the plots demonstrated in Figure 6 show the improvement percentage in RMAE and RRMSE for different initial sample sizes relative to the results of 50% of the total number of sample points. In Figure 6, a negative value indicates that the local performances/overall performances under this initial sample size are inferior to those under 50% of the total number of sample points. The larger the relative improvement percentage value, the higher the degree of excellence.

It is illustrated in Figure 6 that the initial sample size taking half of the total number of sample points adopted in this work is proper for these numerical cases. In addition, it is observed that for these eight numerical examples, too small initial sample size is not enough for get a full knowledge of the discrepancy between HF and LF model, hence, leading some DURs not to be discovered. While too large initial sample size decreases

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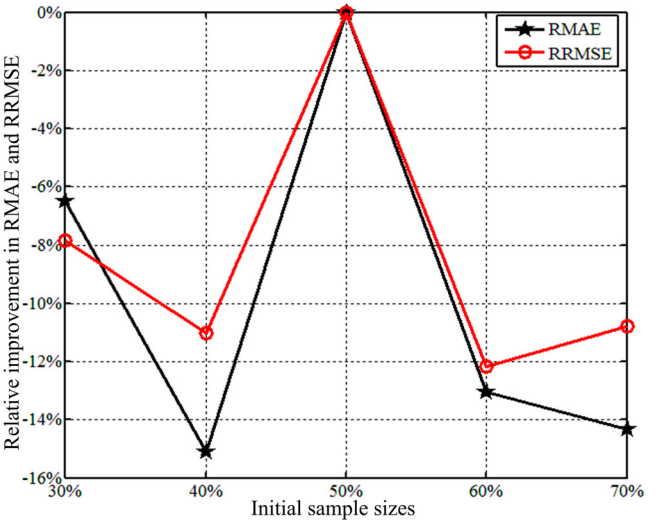


Figure 6. Sensitivity analysis of the initial sample sizes.

the number of sample points for adaptively sampling process, also leading some DURs not to be sampled.

4.6. Sensitivity analysis of the value of the tuning parameter in the space-filling criterion

In this subsection, the sensitivity analysis of the tuning parameter β in space-filling criterion is conducted. Four different tuning parameters values 0.2, 0.3, 0.4, and 0.5 are considered for the Booth (group 1) and Rastrigin's (group 2) functions. The effect of tuning parameter β on the accuracy of the proposed adaptive VF metamodelling approach is illustrated in Figure 7. As can be seen in Figure 7 that the impact of the tuning parameter is not so significant in these cases when $\beta \in [0.2, 0.5]$.

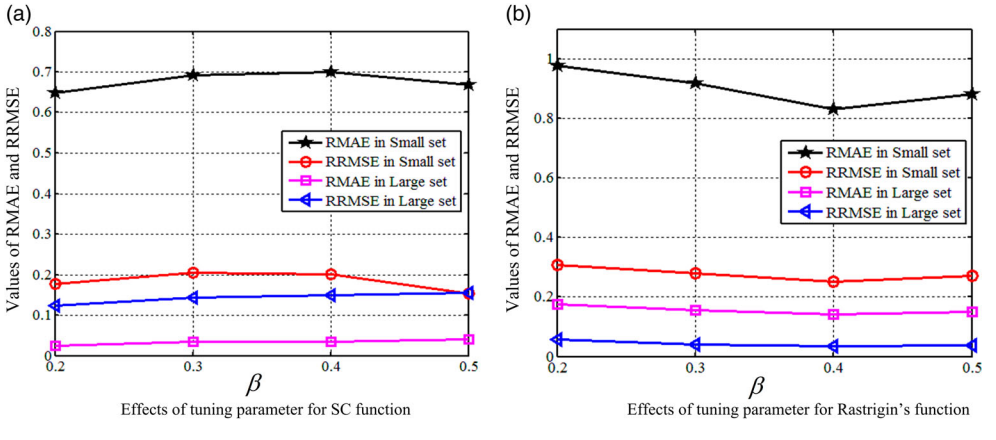


Figure 7. Sensitivity analysis of the tuning parameter.

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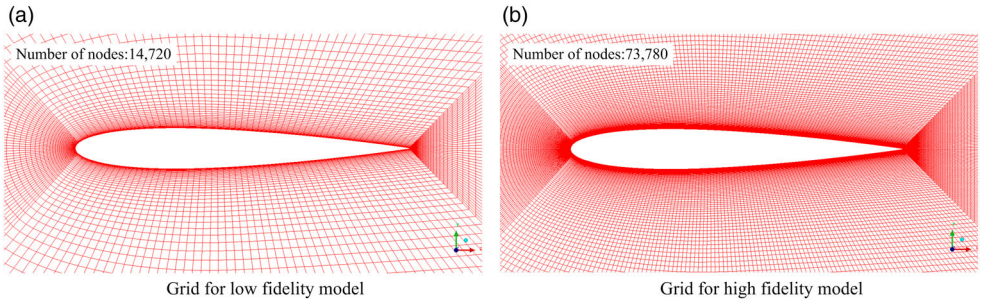


Figure 8. Close up view of the LF and HF CFD grids used in NACA 0012 airfoil.

5. Modelling aerodynamic data for NACA 0012 airfoil

In this section, we consider a NACA 0012 airfoil and use the proposed OSS-EMs-VF meta-modelling approach to generate VFM models for the lift coefficient (C_L) as a function of two independent variables: Mach number Ma and angle of attack α. The ranges for the variables is $0.3 \leq \text{Ma} \leq 0.5$, $0^0 \leq \alpha \leq 10^0$. Two different fidelity models are adopted to calculate C_L. The grid of LF model consists of about 15,000 elements, shown in the Figure 8(a), which extended to 15 times the airfoil length in each direction. For the HF model, the grid consists of about 74,000 elements, shown in the Figure 8(b). Both two fidelity models are solved using Fluent14.5 simulation tool.

Table 3. Sample points and corresponding response values for LF model.

No.	α (*degree*)	Ma	C_L	No.	α (*degree*)	Ma	C_L
1	0.34	0.40	0.0179	16	9.66	0.42	0.4452
2	3.10	0.36	0.1243	17	3.79	0.32	0.1176
3	2.76	0.47	0.1975	18	7.24	0.44	0.4080
4	2.41	0.40	0.1220	19	6.21	0.38	0.2641
5	5.17	0.35	0.1892	20	10.00	0.37	0.3508
6	6.55	0.48	0.4547	21	2.07	0.50	0.1715
7	1.72	0.31	0.0511	22	3.45	0.43	0.2013
8	7.59	0.41	0.3642	23	0.69	0.47	0.0506
9	1.03	0.36	0.0423	24	9.31	0.45	0.5021
10	8.28	0.38	0.3312	25	4.48	0.39	0.2087
11	5.52	0.42	0.2951	26	6.90	0.34	0.2286
12	5.86	0.31	0.1643	27	8.62	0.49	0.5734
13	0.00	0.33	0.0002	28	8.97	0.33	0.2616
14	4.83	0.46	0.3190	29	7.93	0.30	0.1973
15	4.14	0.49	0.3185	30	1.38	0.44	0.0866

Table 4. Sample points and corresponding response values for HF model using Kriging-VF RBF-VFM, SVR-VF and EMs-VF.

No.	α (*degree*)	Ma	C_L	No.	α (*degree*)	Ma	C_L
1	5	0.47	0.3608	9	0.00	0.44	0.0003
2	9.29	0.34	0.3040	10	1.43	0.33	0.0494
3	8.57	0.40	0.4022	11	10.00	0.46	0.5720
4	4.29	0.31	0.1274	12	2.86	0.43	0.2492
5	7.14	0.30	0.1907	13	0.71	0.39	0.0510
6	7.86	0.50	0.6031	14	2.14	0.49	0.2517
7	6.43	0.36	0.2547	15	5.71	0.41	0.4353
8	3.57	0.37	0.1553				

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Table 5. Sample points and corresponding response values for HF model using OSS-EMs-VF.

No.	Ma	α (*degree*)	C_L	No.	Ma	α (*degree*)	C_L
1	7.14	0.5	0.5635	9	7.07	0.44	0.4239
2	8.57	0.36	0.3216	10	0.68	0.5	0.0587
3	10.00	0.44	0.5288	11	10	0.5	0.6603
4	1.43	0.33	0.0494	12	10	0.3	0.2375
5	2.86	0.47	0.2109	13	3.76	0.34	0.1365
6	0.00	0.41	0.0002	14	5.06	0.5	0.4192
7	4.29	0.39	0.2073	15	4.07	0.43	0.2439
8	5.71	0.30	0.1560				

Table 6. Validation points and corresponding response values.

No.	Ma	α (*degree*)	C_L	No.	Ma	α (*degree*)	C_L
1	0	0.39	0.0002	9	3.58	0.43	0.2156
2	0.71	0.46	0.0507	10	8.57	0.30	0.2214
3	1.43	0.33	0.0494	11	9.29	0.36	0.3415
4	2.14	0.43	0.1464	12	10	0.49	0.6374
5	2.86	0.34	0.1044	13	3.57	0.50	0.3015
6	0	0.5	0.0004	14	4.29	0.41	0.2312
7	8.33	0.31	0.2315	15	5.00	0.37	0.2143
8	4.77	0.5	0.3970				

Table 7. Accuracy comparison of the aerodynamic example.

Coefficients	Accuracy metrics	OSS-EMs-VF	EMs-VF	Kriging-VF	RBF-VF	SVR-VF
C_L	RMAE	**0.090**	0.149	***0.361***	0.333	0.192
	RRMSE	**0.026**	0.067	***0.200***	0.196	0.086

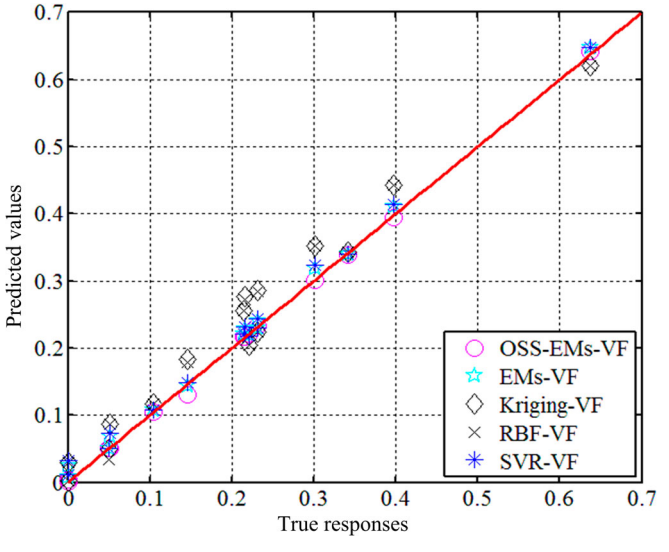


Figure 9. True response and predicted value of the validation points for different VF modelling methods, where the straight line represents that the true response equals to predicted value.

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In this aerodynamic example, 30 sample points are simulated for LF metamodel and the total number of sample points for developing the HF metamodel is limited to 15. For OSS-EMs-VF, an initial design of eight sampling points is generated using OLHS and additional seven sampling points are selected one by one. About 15 validation points are selected randomly to calculate the value of RRMSE and RMAE. The sample points and response values are listed in Tables 3–6.

The comparison results are summarised in Table 7, where the best results for each model type are printed in bold, while the worst results for each model type are printed in bold-italic. It is found that the proposed OSS-EMs-VF provides the most accurate metamodel in which both RRMSE and RMAE are smallest. The accuracy of EMs-VF is somewhere in between OSS-EMs-VF and DMF-SVR. Kriging-VF has a comparable global fitting performance to RBF-VF in this problem, while it has a larger local error. Figure 9 plots the actual simulation values and the corresponding predicted values obtained using different VF metamodel. It is observed that almost all points fall on the straight line for the proposed OSS-EMs-VF approach. (It represents that the true response is equal to predicted value.)

6. Conclusion

In this paper, we propose an active learning global VF metamodelling approach. Since the EMs can take advantage of the prediction ability of each stand-alone metamodel, it is adopted in the proposed OSS-EMs-VF metamodelling approach to map the difference between the HF and LF models on the entire domain. Furthermore, an active learning strategy is introduced in OSS-EMs-VF metamodelling approach aiming to make full use of the already collected experimental data. The proposed approach is compared to four different metamodelling methods (SVR-VF, RBF-VF, Kriging-VF and EMs-VF) using several numerical and engineering cases. The comparison results illustrate that (1) the proposed OSS-EMs-VF approach is distinctly the best in terms of the overall accuracy and robustness; (2) the model fusion technology based on EMs introduced in the proposed metamodelling approach accords with 'insurance policy' mode that it will never be the worst, which can decrease the risk of adopting an inappropriate modelling approach; and (3) the proposed OSS-EMs-VF metamodelling approach, with an active learning strategy, significantly outperforms the EMs-VF metamodelling approach for problems, whose difference response features between HF and LF models are multi-model, non-smooth and have abrupt changes.

It should be pointed out that the proposed OSS-EMs-VF metamodelling method may require additional computational cost to obtain the weight coefficients and solve the objective-oriented sequential sampling problem. However, this additional computational cost is more than likely offset by the savings in calls required for HF models. As part of future work, combining the proposed OSS-EMs-VF metamodelling method with multi-disciplinary optimisation techniques, for example, collaborative optimisation, concurrent subspace optimisation and analytical target cascading, will be investigated. Overall, as a novel VF modelling technique, OSS-EMs-VF exhibits great capability for simulation-based engineering design and optimisation problems.

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Disclosure statement

No potential conflict of interest was reported by the authors.

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